

SOFT TECH

COMPUTER TRAINING INSTITUTE

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Data Science and Data Analysis using Python and Power Bi with Gen AI

3 Months ITR Content Syllabus

Week 1	Python for Data Science (Part 1)	
Python Basics		
Variables, Data Types, <u>Operators</u>		
Control Flow: If-Else, Loops		
Functions & Modules		
List, Tuple, Dictionary, Set		
Object Oriented Programming (OOP)		
Classes & Objects		
Constructors		
Inheritance		
Polymorphism		

Week 2	Python for Data Science (Part 2)	
Exception Handling		
File Handling		
Arrays and Operations		
Indexing, Slicing, Repetition		
Broadcasting		
Python Libraries		
Numpy, Pandas, Matplotlib		
Data Analysis using Pandas		
DataFrames and Series		
Data Cleaning & Preprocessing		

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Week 3	Data Analysis using Python	
Data Visualization using Matplotlib & <u>Seaborn</u>		
Line, Bar, Scatter, Histogram		
<u>Heatmaps, Boxplots</u>		
Explorative Data Analysis (EDA)		
Data Cleaning		
Data Pre-processing		
Data Manipulation		
Missing Value Treatment		
Remove Duplicates		

Week 4	Exploratory Data Analysis (Project)	
Feature Engineering		
Feature Scaling		
Exploratory Data Analysis Project Part 1		
Exploratory Data Analysis Project Part 2		
Project Documentation		
Project Presentation		

Week 5	Introduction to Business Intelligence	
<u>Introduction to Business Intelligence</u>		
<u>Business Data Analysis</u>		
Introduction to Power BI		
Overview of the Power BI Interface		
Understanding Workspaces & Reports		
Basic Charts in Power BI		
Column Chart, Stacked Column Chart, Pie Chart, Donut Chart, Funnel Chart		
Ribbon Charts, Scatter Plots, Box Plots		
<u>Import Visuals, Add other Visuals from Power BI Workspace</u>		

Week 6	Advanced Charts in Power BI Part 1	
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Maps in Power BI
Basic Maps, Filled Maps, Pie Chart Maps
Formatting and Background Customization
<u>Country Specific Maps</u>
Tables and Matrices in Power BI
Creating and Formatting Tables and Matrices, Conditional formatting and Aggregation
Hierarchies, Subtotals, and Totals
Advanced Visualization and <u>Filtering</u>
Waterfall Chart, TreeMap, Gauge Chart
Cards & Filters
Number Cards, Text Cards, Date Cards
Applying Filters to Visuals, Pages and <u>Drillthrough</u>

Week 7	Advanced Charts in Power BI Part 2	
Slicers in Power BI		
Text Slicers, Date Slicers, Number Slicers		
Formatting and Applying Slicers		
Power BI Objects, Bookmarks, Actions		
Insert Images, Text, Shapes and Buttons		
Page Navigation, Bookmarks, and <u>Drillthrough Actions</u>		

Week 8	Power Query Editor	
Introduction to Power Query		
Power Query in Excel and Power BI		
Data Transformation using Power Query		
Text Functions, Date Functions		
Group By, Conditional Columns, etc.		
Appending Data		
Merging Queries		
Column Transformations		

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Week 9	Power Query and DAX	
DAX and Data Modeling, M Language		
Powerful Concepts of Power BI		
Fill Up/Fill Down, Grouping, Transpose, Unpivot		
Data Types, Replace Errors, Remove Rows, Columns		
Introduction to DAX Calculations		
Text functions: UPPER, LOWER, LEFT, RIGHT, CONCATENATE		
Date Functions: Year, Month, Day, Date		
Numerical Functions: SUM, MIN, MAX, AVERAGE, COUNT, SUMX, MINX, MAXX, and AVERAGEX		

Week 10	Power BI (Project)	
Creating or Using Real Data Set		
Creating Attractive Dashboards and Reports		
Effective use of Slicers, Drillthrough and Multipage Navigation		
Creating Measures, Columns, Additional Tables, etc.		
Data Pre-Processing and Cleaning		
Power BI Report Submission		
Project Documentation		
Project Presentation		

Week 11 -12	Generative AI Industry Training	
Working Culture in Companies		
Formal vs Informal Grooming		
Professional Communication		
Industry Relevant Terminologies		
Different AI Tools		
ChatGPT, Gemini, Claude, Eleven Labs		
Midjourney, Meta AI, Copilot		
Chatbot vs AI Model		
Introduction to Technologies like Git, Github, JIRA		